Chapter 7

Functions of Random Variables (Optional)

7.1 Introduction

This chapter contains a broad spectrum of material. Chapters 5 and 6 deal with specific types of distributions, both discrete and continuous. These are distributions that find use in many subject matter applications, including reliability, quality control, and acceptance sampling. In the present chapter, we begin with a more general topic, that of distributions of functions of random variables. General techniques are introduced and illustrated by examples. This discussion is followed by coverage of a related concept, *moment-generating functions*, which can be helpful in learning about distributions of linear functions of random variables.

In standard statistical methods, the result of statistical hypothesis testing, estimation, or even statistical graphics does not involve a single random variable but, rather, *functions of one or more random variables*. As a result, statistical inference requires the distributions of these functions. For example, the use of **averages of random variables** is common. In addition, sums and more general linear combinations are important. We are often interested in the distribution of sums of squares of random variables, particularly in the use of analysis of variance techniques discussed in Chapters 11–14.

7.2 Transformations of Variables

Frequently in statistics, one encounters the need to derive the probability distribution of a function of one or more random variables. For example, suppose that X is a discrete random variable with probability distribution $f(x)$, and suppose further that $Y = u(X)$ defines a one-to-one transformation between the values of X and Y. We wish to find the probability distribution of Y. It is important to note that the one-to-one transformation implies that each value x is related to one, and only one, value $y = u(x)$ and that each value y is related to one, and only one, value $x = w(y)$, where $w(y)$ is obtained by solving $y = u(x)$ for x in terms of y.

From our discussion of discrete probability distributions in Chapter 3, it is clear that the random variable Y assumes the value y when X assumes the value $w(y)$. Consequently, the probability distribution of Y is given by

$$g(y) = P(Y = y) = P[X = w(y)] = f[w(y)].$$

Theorem 7.1: Suppose that X is a **discrete** random variable with probability distribution $f(x)$. Let $Y = u(X)$ define a one-to-one transformation between the values of X and Y so that the equation $y = u(x)$ can be uniquely solved for x in terms of y, say $x = w(y)$. Then the probability distribution of Y is

$$g(y) = f[w(y)].$$

Example 7.1: Let X be a geometric random variable with probability distribution

$$f(x) = \frac{3}{4}\left(\frac{1}{4}\right)^{x-1}, \qquad x = 1, 2, 3, \ldots.$$

Find the probability distribution of the random variable $Y = X^2$.

***Solution*:** Since the values of X are all positive, the transformation defines a one-to-one correspondence between the x and y values, $y = x^2$ and $x = \sqrt{y}$. Hence

$$g(y) = \begin{cases} f(\sqrt{y}) = \frac{3}{4}\left(\frac{1}{4}\right)^{\sqrt{y}-1}, & y = 1, 4, 9, \ldots, \\ 0, & \text{elsewhere.} \end{cases}$$

Similarly, for a two-dimension transformation, we have the result in Theorem 7.2.

Theorem 7.2: Suppose that X_1 and X_2 are **discrete** random variables with joint probability distribution $f(x_1, x_2)$. Let $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ define a one-to-one transformation between the points (x_1, x_2) and (y_1, y_2) so that the equations

$$y_1 = u_1(x_1, x_2) \qquad \text{and} \qquad y_2 = u_2(x_1, x_2)$$

may be uniquely solved for x_1 and x_2 in terms of y_1 and y_2, say $x_1 = w_1(y_1, y_2)$ and $x_2 = w_2(y_1, y_2)$. Then the joint probability distribution of Y_1 and Y_2 is

$$g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)].$$

Theorem 7.2 is extremely useful for finding the distribution of some random variable $Y_1 = u_1(X_1, X_2)$, where X_1 and X_2 are discrete random variables with joint probability distribution $f(x_1, x_2)$. We simply define a second function, say $Y_2 = u_2(X_1, X_2)$, maintaining a one-to-one correspondence between the points (x_1, x_2) and (y_1, y_2), and obtain the joint probability distribution $g(y_1, y_2)$. The distribution of Y_1 is just the marginal distribution of $g(y_1, y_2)$, found by summing over the y_2 values. Denoting the distribution of Y_1 by $h(y_1)$, we can then write

$$h(y_1) = \sum_{y_2} g(y_1, y_2).$$

Example 7.2: Let X_1 and X_2 be two independent random variables having Poisson distributions with parameters μ_1 and μ_2, respectively. Find the distribution of the random variable $Y_1 = X_1 + X_2$.

***Solution*:** Since X_1 and X_2 are independent, we can write

$$f(x_1, x_2) = f(x_1)f(x_2) = \frac{e^{-\mu_1}\mu_1^{x_1}}{x_1!}\frac{e^{-\mu_2}\mu_2^{x_2}}{x_2!} = \frac{e^{-(\mu_1+\mu_2)}\mu_1^{x_1}\mu_2^{x_2}}{x_1!x_2!},$$

where $x_1 = 0, 1, 2, \ldots$ and $x_2 = 0, 1, 2, \ldots$. Let us now define a second random variable, say $Y_2 = X_2$. The inverse functions are given by $x_1 = y_1 - y_2$ and $x_2 = y_2$. Using Theorem 7.2, we find the joint probability distribution of Y_1 and Y_2 to be

$$g(y_1, y_2) = \frac{e^{-(\mu_1+\mu_2)}\mu_1^{y_1-y_2}\mu_2^{y_2}}{(y_1 - y_2)!y_2!},$$

where $y_1 = 0, 1, 2, \ldots$ and $y_2 = 0, 1, 2, \ldots, y_1$. Note that since $x_1 > 0$, the transformation $x_1 = y_1 - x_2$ implies that y_2 and hence x_2 must always be less than or equal to y_1. Consequently, the marginal probability distribution of Y_1 is

$$\begin{aligned} h(y_1) &= \sum_{y_2=0}^{y_1} g(y_1, y_2) = e^{-(\mu_1+\mu_2)} \sum_{y_2=0}^{y_1} \frac{\mu_1^{y_1-y_2}\mu_2^{y_2}}{(y_1 - y_2)!y_2!} \\ &= \frac{e^{-(\mu_1+\mu_2)}}{y_1!} \sum_{y_2=0}^{y_1} \frac{y_1!}{y_2!(y_1 - y_2)!}\mu_1^{y_1-y_2}\mu_2^{y_2} \\ &= \frac{e^{-(\mu_1+\mu_2)}}{y_1!} \sum_{y_2=0}^{y_1} \binom{y_1}{y_2}\mu_1^{y_1-y_2}\mu_2^{y_2}. \end{aligned}$$

Recognizing this sum as the binomial expansion of $(\mu_1 + \mu_2)^{y_1}$ we obtain

$$h(y_1) = \frac{e^{-(\mu_1+\mu_2)}(\mu_1 + \mu_2)^{y_1}}{y_1!}, \qquad y_1 = 0, 1, 2, \ldots,$$

from which we conclude that the sum of the two independent random variables having Poisson distributions, with parameters μ_1 and μ_2, has a Poisson distribution with parameter $\mu_1 + \mu_2$.

To find the probability distribution of the random variable $Y = u(X)$ when X is a continuous random variable and the transformation is one-to-one, we shall need Theorem 7.3. The proof of the theorem is left to the reader.

Theorem 7.3: Suppose that X is a **continuous** random variable with probability distribution $f(x)$. Let $Y = u(X)$ define a one-to-one correspondence between the values of X and Y so that the equation $y = u(x)$ can be uniquely solved for x in terms of y, say $x = w(y)$. Then the probability distribution of Y is

$$g(y) = f[w(y)]|J|,$$

where $J = w'(y)$ and is called the **Jacobian** of the transformation.

Example 7.3: Let X be a continuous random variable with probability distribution

$$f(x) = \begin{cases} \frac{x}{12}, & 1 < x < 5, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability distribution of the random variable $Y = 2X - 3$.

***Solution*:** The inverse solution of $y = 2x - 3$ yields $x = (y + 3)/2$, from which we obtain $J = w'(y) = dx/dy = 1/2$. Therefore, using Theorem 7.3, we find the density function of Y to be

$$g(y) = \begin{cases} \frac{(y+3)/2}{12}\left(\frac{1}{2}\right) = \frac{y+3}{48}, & -1 < y < 7, \\ 0, & \text{elsewhere.} \end{cases}$$

To find the joint probability distribution of the random variables $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ when X_1 and X_2 are continuous and the transformation is one-to-one, we need an additional theorem, analogous to Theorem 7.2, which we state without proof.

Theorem 7.4: Suppose that X_1 and X_2 are **continuous** random variables with joint probability distribution $f(x_1, x_2)$. Let $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ define a one-to-one transformation between the points (x_1, x_2) and (y_1, y_2) so that the equations $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ may be uniquely solved for x_1 and x_2 in terms of y_1 and y_2, say $x_1 = w_1(y_l, y_2)$ and $x_2 = w_2(y_1, y_2)$. Then the joint probability distribution of Y_1 and Y_2 is

$$g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)]|J|,$$

where the Jacobian is the 2×2 determinant

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

and $\frac{\partial x_1}{\partial y_1}$ is simply the derivative of $x_1 = w_1(y_1, y_2)$ with respect to y_1 with y_2 held constant, referred to in calculus as the partial derivative of x_1 with respect to y_1. The other partial derivatives are defined in a similar manner.

Example 7.4: Let X_1 and X_2 be two continuous random variables with joint probability distribution

$$f(x_1, x_2) = \begin{cases} 4x_1x_2, & 0 < x_1 < 1,\ 0 < x_2 < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the joint probability distribution of $Y_1 = X_1^2$ and $Y_2 = X_1X_2$.

***Solution*:** The inverse solutions of $y_1 = x_1^2$ and $y_2 = x_1x_2$ are $x_1 = \sqrt{y_1}$ and $x_2 = y_2/\sqrt{y_1}$, from which we obtain

$$J = \begin{vmatrix} 1/(2\sqrt{y_1}) & 0 \\ -y_2/2y_1^{3/2} & 1/\sqrt{y_1} \end{vmatrix} = \frac{1}{2y_1}.$$

To determine the set B of points in the y_1y_2 plane into which the set A of points in the x_1x_2 plane is mapped, we write

$$x_1 = \sqrt{y_1} \qquad \text{and} \qquad x_2 = y_2/\sqrt{y_1}.$$

Then setting $x_1 = 0$, $x_2 = 0$, $x_1 = 1$, and $x_2 = 1$, the boundaries of set A are transformed to $y_1 = 0$, $y_2 = 0$, $y_1 = 1$, and $y_2 = \sqrt{y_1}$, or $y_2^2 = y_1$. The two regions are illustrated in Figure 7.1. Clearly, the transformation is one-to-one, mapping the set $A = \{(x_1, x_2) \mid 0 < x_1 < 1,\ 0 < x_2 < 1\}$ into the set $B = \{(y_1, y_2) \mid y_2^2 < y_1 < 1,\ 0 < y_2 < 1\}$. From Theorem 7.4 the joint probability distribution of Y_1 and Y_2 is

$$g(y_1, y_2) = 4(\sqrt{y_1})\frac{y_2}{\sqrt{y_1}}\frac{1}{2y_1} = \begin{cases} \frac{2y_2}{y_1}, & y_2^2 < y_1 < 1,\ 0 < y_2 < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

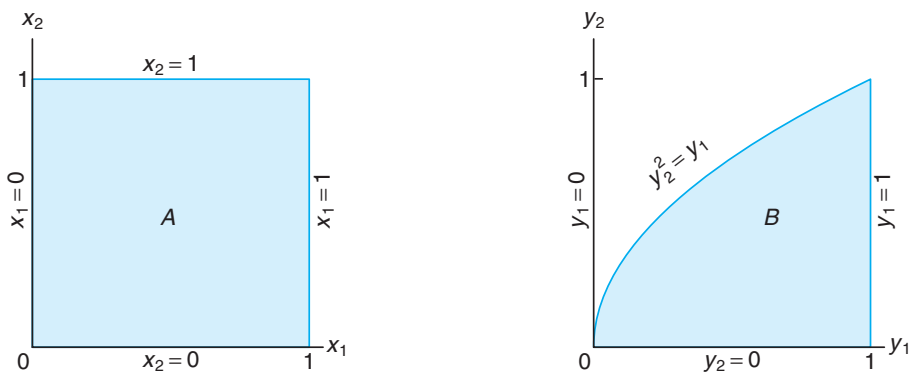

Figure 7.1: Mapping set A into set B.

Problems frequently arise when we wish to find the probability distribution of the random variable $Y = u(X)$ when X is a continuous random variable and the transformation is not one-to-one. That is, to each value x there corresponds exactly one value y, but to each y value there corresponds more than one x value. For example, suppose that $f(x)$ is positive over the interval $-1 < x < 2$ and zero elsewhere. Consider the transformation $y = x^2$. In this case, $x = \pm\sqrt{y}$ for $0 < y < 1$ and $x = \sqrt{y}$ for $1 < y < 4$. For the interval $1 < y < 4$, the probability distribution of Y is found as before, using Theorem 7.3. That is,

$$g(y) = f[w(y)]|J| = \frac{f(\sqrt{y})}{2\sqrt{y}}, \qquad 1 < y < 4.$$

However, when $0 < y < 1$, we may partition the interval $-1 < x < 1$ to obtain the two inverse functions

$$x = -\sqrt{y}, \qquad -1 < x < 0, \qquad \text{and} \qquad x = \sqrt{y}, \qquad 0 < x < 1.$$

Then to every y value there corresponds a single x value for each partition. From Figure 7.2 we see that

$$\begin{aligned} P(a < Y < b) &= P(-\sqrt{b} < X < -\sqrt{a}) + P(\sqrt{a} < X < \sqrt{b}) \\ &= \int_{-\sqrt{b}}^{-\sqrt{a}} f(x)\ dx + \int_{\sqrt{a}}^{\sqrt{b}} f(x)\ dx. \end{aligned}$$

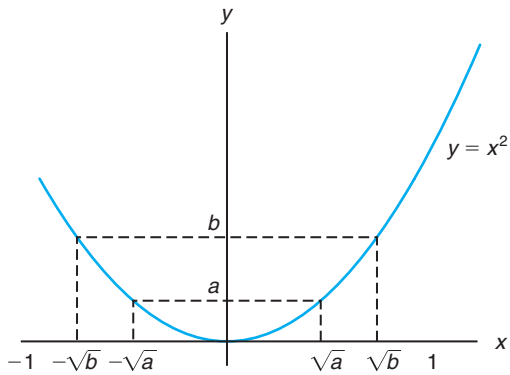

Figure 7.2: Decreasing and increasing function.

Changing the variable of integration from x to y, we obtain

$$\begin{aligned} P(a < Y < b) &= \int_b^a f(-\sqrt{y})J_1\ dy + \int_a^b f(\sqrt{y})J_2\ dy \\ &= -\int_a^b f(-\sqrt{y})J_1\ dy + \int_a^b f(\sqrt{y})J_2\ dy, \end{aligned}$$

where

$$J_1 = \frac{d(-\sqrt{y})}{dy} = \frac{-1}{2\sqrt{y}} = -|J_1|$$

and

$$J_2 = \frac{d(\sqrt{y})}{dy} = \frac{1}{2\sqrt{y}} = |J_2|.$$

Hence, we can write

$$P(a < Y < b) = \int_a^b [f(-\sqrt{y})|J_1| + f(\sqrt{y})|J_2|]\ dy,$$

and then

$$g(y) = f(-\sqrt{y})|J_1| + f(\sqrt{y})|J_2| = \frac{f(-\sqrt{y}) + f(\sqrt{y})}{2\sqrt{y}}, \qquad 0 < y < 1.$$

The probability distribution of Y for $0 < y < 4$ may now be written

$$g(y) = \begin{cases} \frac{f(-\sqrt{y})+f(\sqrt{y})}{2\sqrt{y}}, & 0 < y < 1, \\ \frac{f(\sqrt{y})}{2\sqrt{y}}, & 1 < y < 4, \\ 0, & \text{elsewhere.} \end{cases}$$

This procedure for finding $g(y)$ when $0 < y < 1$ is generalized in Theorem 7.5 for k inverse functions. For transformations not one-to-one of functions of several variables, the reader is referred to *Introduction to Mathematical Statistics* by Hogg, McKean, and Craig (2005; see the Bibliography).

Theorem 7.5: Suppose that X is a **continuous** random variable with probability distribution $f(x)$. Let $Y = u(X)$ define a transformation between the values of X and Y that is not one-to-one. If the interval over which X is defined can be partitioned into k mutually disjoint sets such that each of the inverse functions

$$x_1 = w_1(y), \quad x_2 = w_2(y), \quad \ldots, \quad x_k = w_k(y)$$

of $y = u(x)$ defines a one-to-one correspondence, then the probability distribution of Y is

$$g(y) = \sum_{i=1}^{k} f[w_i(y)]|J_i|,$$

where $J_i = w_i^{'}(y)$, $i = 1, 2, \ldots, k$.

Example 7.5: Show that $Y = (X-\mu)^2/\sigma^2$ has a chi-squared distribution with 1 degree of freedom when X has a normal distribution with mean μ and variance σ^2.

***Solution*:** Let $Z = (X - \mu)/\sigma$, where the random variable Z has the standard normal distribution

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad -\infty < z < \infty.$$

We shall now find the distribution of the random variable $Y = Z^2$. The inverse solutions of $y = z^2$ are $z = \pm\sqrt{y}$. If we designate $z_1 = -\sqrt{y}$ and $z_2 = \sqrt{y}$, then $J_1 = -1/2\sqrt{y}$ and $J_2 = 1/2\sqrt{y}$. Hence, by Theorem 7.5, we have

$$g(y) = \frac{1}{\sqrt{2\pi}} e^{-y/2} \left| \frac{-1}{2\sqrt{y}} \right| + \frac{1}{\sqrt{2\pi}} e^{-y/2} \left| \frac{1}{2\sqrt{y}} \right| = \frac{1}{\sqrt{2\pi}} y^{1/2-1} e^{-y/2}, \qquad y > 0.$$

Since $g(y)$ is a density function, it follows that

$$1 = \frac{1}{\sqrt{2\pi}} \int_0^\infty y^{1/2-1} e^{-y/2}\, dy = \frac{\Gamma(1/2)}{\sqrt{\pi}} \int_0^\infty \frac{y^{1/2-1} e^{-y/2}}{\sqrt{2}\Gamma(1/2)}\, dy = \frac{\Gamma(1/2)}{\sqrt{\pi}},$$

the integral being the area under a gamma probability curve with parameters $\alpha = 1/2$ and $\beta = 2$. Hence, $\sqrt{\pi} = \Gamma(1/2)$ and the density of Y is given by

$$g(y) = \begin{cases} \frac{1}{\sqrt{2}\Gamma(1/2)} y^{1/2-1} e^{-y/2}, & y > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

which is seen to be a chi-squared distribution with 1 degree of freedom. ∎

7.3 Moments and Moment-Generating Functions

In this section, we concentrate on applications of moment-generating functions. The obvious purpose of the moment-generating function is in determining moments of random variables. However, the most important contribution is to establish distributions of functions of random variables.

If $g(X) = X^r$ for $r = 0, 1, 2, 3, \ldots$, Definition 7.1 yields an expected value called the rth **moment about the origin** of the random variable X, which we denote by μ'_r.

Definition 7.1: The rth **moment about the origin** of the random variable X is given by

$$\mu'_r = E(X^r) = \begin{cases} \sum\limits_x x^r f(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} x^r f(x)\, dx, & \text{if } X \text{ is continuous.} \end{cases}$$

Since the first and second moments about the origin are given by $\mu'_1 = E(X)$ and $\mu'_2 = E(X^2)$, we can write the mean and variance of a random variable as

$$\mu = \mu'_1 \qquad \text{and} \qquad \sigma^2 = \mu'_2 - \mu^2.$$

Although the moments of a random variable can be determined directly from Definition 7.1, an alternative procedure exists. This procedure requires us to utilize a **moment-generating function**.

Definition 7.2: The **moment-generating function** of the random variable X is given by $E(e^{tX})$ and is denoted by $M_X(t)$. Hence,

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum\limits_x e^{tx} f(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\, dx, & \text{if } X \text{ is continuous.} \end{cases}$$

Moment-generating functions will exist only if the sum or integral of Definition 7.2 converges. If a moment-generating function of a random variable X does exist, it can be used to generate all the moments of that variable. The method is described in Theorem 7.6 without proof.

Theorem 7.6: Let X be a random variable with moment-generating function $M_X(t)$. Then

$$\left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} = \mu'_r.$$

Example 7.6: Find the moment-generating function of the binomial random variable X and then use it to verify that $\mu = np$ and $\sigma^2 = npq$.

***Solution*:** From Definition 7.2 we have

$$M_X(t) = \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x q^{n-x} = \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x q^{n-x}.$$

Recognizing this last sum as the binomial expansion of $(pe^t + q)^n$, we obtain

$$M_X(t) = (pe^t + q)^n.$$

Now

$$\frac{dM_X(t)}{dt} = n(pe^t + q)^{n-1}pe^t$$

and

$$\frac{d^2M_X(t)}{dt^2} = np[e^t(n-1)(pe^t + q)^{n-2}pe^t + (pe^t + q)^{n-1}e^t].$$

Setting $t = 0$, we get

$$\mu_1' = np \text{ and } \mu_2' = np[(n-1)p + 1].$$

Therefore,

$$\mu = \mu_1' = np \text{ and } \sigma^2 = \mu_2' - \mu^2 = np(1-p) = npq,$$

which agrees with the results obtained in Chapter 5.

Example 7.7: Show that the moment-generating function of the random variable X having a normal probability distribution with mean μ and variance σ^2 is given by

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$

***Solution*:** From Definition 7.2 the moment-generating function of the normal random variable X is

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{x^2 - 2(\mu + t\sigma^2)x + \mu^2}{2\sigma^2}\right] dx. \end{aligned}$$

Completing the square in the exponent, we can write

$$x^2 - 2(\mu + t\sigma^2)x + \mu^2 = [x - (\mu + t\sigma^2)]^2 - 2\mu t\sigma^2 - t^2\sigma^4$$

and then

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{[x - (\mu + t\sigma^2)]^2 - 2\mu t\sigma^2 - t^2\sigma^4}{2\sigma^2}\right\} dx \\ &= \exp\left(\frac{2\mu t + \sigma^2 t^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{[x - (\mu + t\sigma^2)]^2}{2\sigma^2}\right\} dx. \end{aligned}$$

Let $w = [x - (\mu + t\sigma^2)]/\sigma$; then $dx = \sigma\, dw$ and

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right),$$

since the last integral represents the area under a standard normal density curve and hence equals 1.

Although the method of transforming variables provides an effective way of finding the distribution of a function of several variables, there is an alternative and often preferred procedure when the function in question is a linear combination of independent random variables. This procedure utilizes the properties of moment-generating functions discussed in the following four theorems. In keeping with the mathematical scope of this book, we state Theorem 7.7 without proof.

Theorem 7.7: **(Uniqueness Theorem)** Let X and Y be two random variables with moment-generating functions $M_X(t)$ and $M_Y(t)$, respectively. If $M_X(t) = M_Y(t)$ for all values of t, then X and Y have the same probability distribution.

Theorem 7.8: $M_{X+a}(t) = e^{at}M_X(t)$.

***Proof*:** $M_{X+a}(t) = E[e^{t(X+a)}] = e^{at}E(e^{tX}) = e^{at}M_X(t)$.

Theorem 7.9: $M_{aX}(t) = M_X(at)$.

***Proof*:** $M_{aX}(t) = E[e^{t(aX)}] = E[e^{(at)X}] = M_X(at)$.

Theorem 7.10: If $X_1, X_2, \ldots, X_n$ are independent random variables with moment-generating functions $M_{X_1}(t), M_{X_2}(t), \ldots, M_{X_n}(t)$, respectively, and $Y = X_1 + X_2 + \cdots + X_n$, then

$$M_Y(t) = M_{X_1}(t)M_{X_2}(t)\cdots M_{X_n}(t).$$

The proof of Theorem 7.10 is left for the reader.

Theorems 7.7 through 7.10 are vital for understanding moment-generating functions. An example follows to illustrate. There are many situations in which we need to know the distribution of the sum of random variables. We may use Theorems 7.7 and 7.10 and the result of Exercise 7.19 on page 224 to find the distribution of a sum of two independent Poisson random variables with moment-generating functions given by

$$M_{X_1}(t) = e^{\mu_1(e^t-1)} \text{ and } M_{X_2}(t) = e^{\mu_2(e^t-1)},$$

respectively. According to Theorem 7.10, the moment-generating function of the random variable $Y_1 = X_1 + X_2$ is

$$M_{Y_1}(t) = M_{X_1}(t)M_{X_2}(t) = e^{\mu_1(e^t-1)}e^{\mu_2(e^t-1)} = e^{(\mu_1+\mu_2)(e^t-1)},$$

which we immediately identify as the moment-generating function of a random variable having a Poisson distribution with the parameter $\mu_1 + \mu_2$. Hence, according to Theorem 7.7, we again conclude that the sum of two independent random variables having Poisson distributions, with parameters μ_1 and μ_2, has a Poisson distribution with parameter $\mu_1 + \mu_2$.

Linear Combinations of Random Variables

In applied statistics one frequently needs to know the probability distribution of a linear combination of independent normal random variables. Let us obtain the distribution of the random variable $Y = a_1X_1 + a_2X_2$ when X_1 is a normal variable with mean μ_1 and variance σ_1^2 and X_2 is also a normal variable but independent of X_1 with mean μ_2 and variance σ_2^2. First, by Theorem 7.10, we find

$$M_Y(t) = M_{a_1X_1}(t)M_{a_2X_2}(t),$$

and then, using Theorem 7.9, we find

$$M_Y(t) = M_{X_1}(a_1t)M_{X_2}(a_2t).$$

Substituting a_1t for t and then a_2t for t in a moment-generating function of the normal distribution derived in Example 7.7, we have

$$\begin{aligned} M_Y(t) &= \exp(a_1\mu_1t + a_1^2\sigma_1^2t^2/2 + a_2\mu_2t + a_2^2\sigma_2^2t^2/2) \\ &= \exp[(a_1\mu_1 + a_2\mu_2)t + (a_1^2\sigma_1^2 + a_2^2\sigma_2^2)t^2/2], \end{aligned}$$

which we recognize as the moment-generating function of a distribution that is normal with mean $a_1\mu_1 + a_2\mu_2$ and variance $a_1^2\sigma_1^2 + a_2^2\sigma_2^2$.

Generalizing to the case of n independent normal variables, we state the following result.

Theorem 7.11: If $X_1, X_2, \ldots, X_n$ are independent random variables having normal distributions with means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, respectively, then the random variable

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n$$

has a normal distribution with mean

$$\mu_Y = a_1\mu_1 + a_2\mu_2 + \cdots + a_n\mu_n$$

and variance

$$\sigma_Y^2 = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \cdots + a_n^2\sigma_n^2.$$

It is now evident that the Poisson distribution and the normal distribution possess a reproductive property in that the sum of independent random variables having either of these distributions is a random variable that also has the same type of distribution. The chi-squared distribution also has this reproductive property.

Theorem 7.12: If $X_1, X_2, \ldots, X_n$ are mutually independent random variables that have, respectively, chi-squared distributions with $v_1, v_2, \ldots, v_n$ degrees of freedom, then the random variable

$$Y = X_1 + X_2 + \cdots + X_n$$

has a chi-squared distribution with $v = v_1 + v_2 + \cdots + v_n$ degrees of freedom.

***Proof*:** By Theorem 7.10 and Exercise 7.21,

$$M_Y(t) = M_{X_1}(t)M_{X_2}(t)\cdots M_{X_n}(t) \text{ and } M_{X_i}(t) = (1-2t)^{-v_i/2},\ i = 1, 2, \ldots, n.$$

Therefore,

$$M_Y(t) = (1-2t)^{-v_1/2}(1-2t)^{-v_2/2}\cdots(1-2t)^{-v_n/2} = (1-2t)^{-(v_1+v_2+\cdots+v_n)/2},$$

which we recognize as the moment-generating function of a chi-squared distribution with $v = v_1 + v_2 + \cdots + v_n$ degrees of freedom.

Corollary 7.1: If $X_1, X_2, \ldots, X_n$ are independent random variables having identical normal distributions with mean μ and variance σ^2, then the random variable

$$Y = \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2$$

has a chi-squared distribution with $v = n$ degrees of freedom.

This corollary is an immediate consequence of Example 7.5. It establishes a relationship between the very important chi-squared distribution and the normal distribution. It also should provide the reader with a clear idea of what we mean by the parameter that we call degrees of freedom. In future chapters, the notion of degrees of freedom will play an increasingly important role.

Corollary 7.2: If $X_1, X_2, \ldots, X_n$ are independent random variables and X_i follows a normal distribution with mean μ_i and variance σ_i^2 for $i = 1, 2, \ldots, n$, then the random variable

$$Y = \sum_{i=1}^{n} \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2$$

has a chi-squared distribution with $v = n$ degrees of freedom.

Exercises

7.1 Let X be a random variable with probability

$$f(x) = \begin{cases} \frac{1}{3}, & x = 1, 2, 3, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability distribution of the random variable $Y = 2X - 1$.

7.2 Let X be a binomial random variable with probability distribution

$$f(x) = \begin{cases} \binom{3}{x}\left(\frac{2}{5}\right)^x\left(\frac{3}{5}\right)^{3-x}, & x = 0, 1, 2, 3, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability distribution of the random variable $Y = X^2$.

7.3 Let X_1 and X_2 be discrete random variables with the joint multinomial distribution

$$f(x_1, x_2) = \binom{2}{x_1, x_2, 2 - x_1 - x_2}\left(\frac{1}{4}\right)^{x_1}\left(\frac{1}{3}\right)^{x_2}\left(\frac{5}{12}\right)^{2-x_1-x_2}$$

for $x_1 = 0, 1, 2$; $x_2 = 0, 1, 2$; $x_1 + x_2 \leq 2$; and zero elsewhere. Find the joint probability distribution of $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$.

7.4 Let X_1 and X_2 be discrete random variables with joint probability distribution

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{18}, & x_1 = 1, 2;\ x_2 = 1, 2, 3, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability distribution of the random variable $Y = X_1 X_2$.

7.5 Let X have the probability distribution

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that the random variable $Y = -2 \ln X$ has a chi-squared distribution with 2 degrees of freedom.

7.6 Given the random variable X with probability distribution

$$f(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find the probability distribution of $Y = 8X^3$.

7.7 The speed of a molecule in a uniform gas at equilibrium is a random variable V whose probability distribution is given by

$$f(v) = \begin{cases} kv^2 e^{-bv^2}, & v > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where k is an appropriate constant and b depends on the absolute temperature and mass of the molecule. Find the probability distribution of the kinetic energy of the molecule W, where $W = mV^2/2$.

7.8 A dealer's profit, in units of \$5000, on a new automobile is given by $Y = X^2$, where X is a random variable having the density function

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the probability density function of the random variable Y.

(b) Using the density function of Y, find the probability that the profit on the next new automobile sold by this dealership will be less than \$500.

7.9 The hospital period, in days, for patients following treatment for a certain type of kidney disorder is a random variable $Y = X + 4$, where X has the density function

$$f(x) = \begin{cases} \frac{32}{(x+4)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the probability density function of the random variable Y.

(b) Using the density function of Y, find the probability that the hospital period for a patient following this treatment will exceed 8 days.

7.10 The random variables X and Y, representing the weights of creams and toffees, respectively, in 1-kilogram boxes of chocolates containing a mixture of creams, toffees, and cordials, have the joint density function

$$f(x,y) = \begin{cases} 24xy, & 0 \le x \le 1,\ 0 \le y \le 1, x + y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the probability density function of the random variable $Z = X + Y$.

(b) Using the density function of Z, find the probability that, in a given box, the sum of the weights of creams and toffees accounts for at least 1/2 but less than 3/4 of the total weight.

7.11 The amount of kerosene, in thousands of liters, in a tank at the beginning of any day is a random amount Y from which a random amount X is sold during that day. Assume that the joint density function of these variables is given by

$$f(x,y) = \begin{cases} 2, & 0 < x < y,\ 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability density function for the amount of kerosene left in the tank at the end of the day.

7.12 Let X_1 and X_2 be independent random variables each having the probability distribution

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that the random variables Y_1 and Y_2 are independent when $Y_1 = X_1 + X_2$ and $Y_2 = X_1/(X_1 + X_2)$.

7.13 A current of I amperes flowing through a resistance of R ohms varies according to the probability distribution

$$f(i) = \begin{cases} 6i(1-i), & 0 < i < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

If the resistance varies independently of the current according to the probability distribution

$$g(r) = \begin{cases} 2r, & 0 < r < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

find the probability distribution for the power $W = I^2 R$ watts.

7.14 Let X be a random variable with probability distribution

$$f(x) = \begin{cases} \frac{1+x}{2}, & -1 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability distribution of the random variable $Y = X^2$.

7.15 Let X have the probability distribution

$$f(x) = \begin{cases} \frac{2(x+1)}{9}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability distribution of the random variable $Y = X^2$.

7.16 Show that the rth moment about the origin of the gamma distribution is

$$\mu_r' = \frac{\beta^r \Gamma(\alpha + r)}{\Gamma(\alpha)}.$$

[*Hint*: Substitute $y = x/\beta$ in the integral defining μ_r' and then use the gamma function to evaluate the integral.]

7.17 A random variable X has the discrete uniform distribution

$$f(x; k) = \begin{cases} \frac{1}{k}, & x = 1, 2, \ldots, k, \\ 0, & \text{elsewhere.} \end{cases}$$

Show that the moment-generating function of X is

$$M_X(t) = \frac{e^t(1 - e^{kt})}{k(1 - e^t)}.$$

7.18 A random variable X has the geometric distribution $g(x; p) = pq^{x-1}$ for $x = 1, 2, 3, \ldots$. Show that the moment-generating function of X is

$$M_X(t) = \frac{pe^t}{1 - qe^t}, \qquad t < \ln q,$$

and then use $M_X(t)$ to find the mean and variance of the geometric distribution.

7.19 A random variable X has the Poisson distribution $p(x; \mu) = e^{-\mu}\mu^x/x!$ for $x = 0, 1, 2, \ldots$. Show that the moment-generating function of X is

$$M_X(t) = e^{\mu(e^t - 1)}.$$

Using $M_X(t)$, find the mean and variance of the Poisson distribution.

7.20 The moment-generating function of a certain Poisson random variable X is given by

$$M_X(t) = e^{4(e^t - 1)}.$$

Find $P(\mu - 2\sigma < X < \mu + 2\sigma)$.

7.21 Show that the moment-generating function of the random variable X having a chi-squared distribution with v degrees of freedom is

$$M_X(t) = (1 - 2t)^{-v/2}.$$

7.22 Using the moment-generating function of Exercise 7.21, show that the mean and variance of the chi-squared distribution with v degrees of freedom are, respectively, v and $2v$.

7.23 If both X and Y, distributed independently, follow exponential distributions with mean parameter 1, find the distributions of

(a) $U = X + Y$;

(b) $V = X/(X + Y)$.

7.24 By expanding e^{tx} in a Maclaurin series and integrating term by term, show that

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f(x)\, dx \\ &= 1 + \mu t + \mu_2' \frac{t^2}{2!} + \cdots + \mu_r' \frac{t^r}{r!} + \cdots . \end{aligned}$$